Log-Prime Encoded Optical Multiplexing and Aggregate Readout

Revised Technical Disclosure

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Important status: This document describes a mathematical coding method and proposed optical implementations. Existing multi-lane results are software simulations, not physically calibrated optical measurements. The 65,536-state 8+8 extension is a theoretical two-observable target, not an experimentally demonstrated capacity. Novelty and patentability have not been legally determined.

Prepared as an expanded replacement for Log_Prime_Optical_Encoding_Revised_2026.pdf

Revision summary

Version 2.1 preserves the corrected receiver physics of v2.0 and adds the detailed finite-precision analysis developed after the original revision. The principal changes are below.

Addition or correction	Version 2.1 treatment
Binary subset-mean uniqueness	Proves that exact arithmetic means of distinct binary prime subsets are unique even when active count is unknown. Distinguishes this codebook result from direct product reconstruction, which still needs total count or weight.
Capacity versus attainability	Separates the 10^d-bin information ceiling from the actual log-prime codebook. A nominal d-decimal measurement does not guarantee that all irregular subset means occupy different bins.
Complete subset enumeration	Defines the 2^n-state codebook, global minimum separation, rounded-bin collision test, and nearest-neighbour analysis.
Worked first-prime results	Adds n=1 to 9 calculations. With naive first primes and three-decimal rounding, n=6 is collision-free in ideal arithmetic; n=7 has collisions. Robust engineering margins are lower.
Scaling A and offset B	Shows that affine scaling changes physical separation but cannot create more normalized detector bins unless the usable range also expands. Offset translates the codebook only.
Noise-aware precision	Relates minimum code distance to ADC step, RMS uncertainty, drift, calibration error, and rejection thresholds.
8+8 orthogonal extension	Defines an 8-bit spectral word and independent 8-bit binary-weighted common-amplitude word. The Cartesian product is 65,536 permitted states only when both observables are independently controllable and resolvable.
Terminology	Uses optical power/intensity for additive detection. Coherent field amplitudes require phase treatment and are not assumed to add as powers.

Abstract

This disclosure describes an optical channel code in which an intended calibrated lane coordinate is generated from a distinct prime p_i according to C_i = A log(p_i) + B. For active binary lanes, the arithmetic mean of the selected log-prime coordinates is mathematically unique for every non-empty subset, even when the number of active lanes is not separately known. This exact uniqueness follows from unique prime factorisation. It does not guarantee distinguishability at finite detector precision.

When total active count or total integer weight is available, multiplying the normalized mean by that count or weight and exponentiating recovers a product of prime powers. Prime support identifies active lanes and integer exponents identify multiplicities. Without count, a finite precomputed mean codebook can still decode binary subsets by nearest valid mean, subject to noise and finite precision.

A physically defensible aggregate receiver is modelled using calibrated observables such as aggregate optical power P and photocurrent I. Their ratio has units of volts and may be transformed into a weighted mean photon-energy or mean-frequency estimate only under stated

calibration assumptions. Hardware decoding should therefore use measured codepoints, covariance estimates, rejection logic, and dark-state handling.

The revision gives a detailed precision framework. An instrument with d decimal places over a normalized unit interval has at most 10^d nominal bins, so binary lane count satisfies 2^n <= 10^d as an information-theoretic ceiling. The actual log-prime codebook is irregular, and practical lane count is governed by its minimum calibrated code distance, rounding collisions, and noise margin. An orthogonal extension combines an 8-bit spectral codebook with an independently controlled 8-bit binary-weighted common intensity scale, yielding up to 65,536 permitted two-observable states in theory.

1. Scope and contribution

The disclosed contribution is the use of prime-derived logarithmic lane coordinates together with aggregate mean, optional total-weight recovery, calibrated codebook decoding, and a proposed independent amplitude dimension. Established components such as wavelength-division multiplexing, filters, gratings, lenses, spatial light modulators, power meters, photodiodes, interferometers, ADCs, and optical gates are not claimed individually.

The disclosure covers free-space and integrated implementations provided the receiver produces calibrated observables equivalent to the required code coordinates. Exact factorisation is retained as an ideal interpretation; practical decoding may instead use nearest-codeword or maximum-likelihood classification.

2. Definitions and notation

Symbol	Definition
p_i	Distinct prime assigned to spectral lane i
C_i	Ideal lane code coordinate A log(p_i) + B
b_i	Binary spectral-lane state, 0 or 1
w_i	Calibrated non-negative lane multiplicity or integer weight
S	Non-empty active binary lane subset
N = |S|	Number of active binary lanes
W	Total integer weight, sum_i w_i
mu(S)	Arithmetic mean of C_i over active subset S
Phi_i	Photon flux at receiver reference plane
eta_i	Effective photon-to-collected-charge efficiency
P	Aggregate optical power
I	Aggregate photocurrent
q	Elementary charge
Delta_min	Minimum distance between distinct valid codepoints
q_ADC	ADC quantisation step in the calibrated coordinate
sigma	RMS aggregate uncertainty after calibration
D	Independent binary-weighted amplitude word, when enabled

3. Ideal log-prime decoding theory

3.1 Equal calibrated binary contributions

$$C_i = A \log(p_i) + B$$

$$\mu(S) = (1/N) \sum_{i \in S} C_i$$

$$M_{log} = (\mu(S) - B) / A = (1/N) \sum_{i \in S} \log(p_i)$$

If N is known, direct product reconstruction is:

$$\text{base}^{(N M_{log})} = \prod_{i \in S} p_i$$

The exponentiation base must match the logarithm base. The factorisation route is exact only in ideal arithmetic and should not be forced when uncertainty permits more than one valid state.

3.2 Integer-weighted contributions

$$M_C = [\sum_i w_i C_i] / W, \quad W = \sum_i w_i$$

$$M_{log} = (M_C - B) / A$$

$$\text{base}^{(W M_{log})} = \prod_i p_i^{(w_i)}$$

Direct factor exponents require non-negative integer w_i. Rational weights may be rescaled only if a common denominator is known. Signed or continuous weights require extra sign channels, a bounded calibrated codebook, or a different decoder.

3.3 Theorem: exact binary subset means are unique without known N

Theorem. For distinct primes p_i and non-empty binary subsets S and T, define $m(S) = |S|^{-1} \sum_{i \in S} \log(p_i)$. Then $m(S) = m(T)$ if and only if $S = T$.

Proof. Let $P_S = \prod_{i \in S} p_i$ and $P_T = \prod_{i \in T} p_i$. If $m(S) = m(T)$, then:

$$\log(P_S) / |S| = \log(P_T) / |T|$$

$$P_S^{|T|} = P_T^{|S|}$$

P_S and P_T are square-free products. In the left-hand side, every prime in S has exponent $|T|$ and every prime outside S has exponent zero. In the right-hand side, every prime in T has exponent $|S|$ and every prime outside T has exponent zero. Unique prime factorisation requires identical prime support, hence $S = T$. The converse is immediate.

Engineering qualification: The theorem proves exact uniqueness, not finite-precision separability. Two distinct means may be arbitrarily close relative to detector uncertainty. Practical feasibility is therefore a code-distance problem.

3.4 Two ideal decoding modes

Mode	Required observations	Operation	Limitation
Factor decoder	Mean plus known N or W	Normalize, multiply by N or W, exponentiate, factorise	Numerically sensitive; requires count/weight and base consistency
Mean-codebook decoder	Mean only for binary subsets	Compare measured mean with all precomputed valid subset means	State count grows as 2^n; finite precision and calibration control performance

Two-observable statistical decoder	Calibrated observation vector such as [P,I]	Nearest-codeword or maximum-likelihood classification	Requires measured means, covariances, and rejection thresholds

4. Corrected aggregate receiver physics

$$P = \sum_i a_i \Phi_i h \nu_i$$

$$I = q \sum_i b_i \eta_i \Phi_i$$

$$P/I = (h/q) \left[\sum_i a_i \Phi_i \nu_i\right] / \left[\sum_i b_i \eta_i \Phi_i\right]$$

P/I is not automatically mean photon energy divided by charge. It becomes proportional to a photon-flux-weighted mean frequency only when the wavelength-dependent path, efficiency, and gain factors are equal or are transformed by calibration to an equivalent common response.

$$P/I = (h / q \eta_{eff}) \left[\sum_i \Phi_i \nu_i\right] / \left[\sum_i \Phi_i\right] \quad \text{calibrated special case}$$

A normal photodiode terminal voltage should not be described as a direct measurement of mean photon energy. A practical decoder may operate on calibrated P and I directly rather than imposing an exact voltage law.

4.1 Equal contribution

- Equal optical power per lane does not imply equal photon flux, because photon energy is h nu.
- Equal photon flux does not imply equal photocurrent when eta_i differs by wavelength.
- The recommended binary definition is equal calibrated contribution at a stated receiver reference plane.

4.2 Dark state

For no active lane, P and I approach background levels and any ratio is undefined or noise dominated. The receiver must first classify the dark state using calibrated thresholds. Dark is the zero spectral word and should not be inserted into the non-empty mean codebook.

5. Target coordinates, physical wavelengths, and calibration

The prime law generates target coordinates. A grating, filter bank, modulator, or pixelated implementation maps those targets to physical channels. Actual centre frequencies, bandwidths, losses, and responsivities differ from the ideal values, so exact factorisation requires correction or the decoder must use a measured codebook.

Coordinate or record	Purpose
Target frequency f_i*	Ideal prime-derived design coordinate
Target wavelength lambda_i* = c/f_i*	Optical design target
Physical lane or pixel	Manufacturable/discrete implementation
Measured centre and bandwidth	Actual spectral coordinate and overlap
Single-lane P and I sweeps	Gain, linearity, uncertainty, and wavelength response
State codebook	Predicted and measured aggregate codepoints

6. Optical-logic qualification

A software rule such as OUT[x] = mapped(x) AND NOT(INPUT1[x] OR INPUT2[x]) is a valid Boolean convention but is not itself a physical optical NOR gate. Passive attenuation cannot create a bright output when both signal inputs are dark. A physical NOR requires a probe or bias field and an active or interferometric mechanism by which either input suppresses, diverts, phase-cancels, or switches the probe.

A physical claim requires a measured four-input truth table, contrast, threshold, extinction ratio, latency, repeatability, fan-out, restoration, and cascadability.

7. Robust decoding procedure

1. Acquire synchronous calibrated observables, for example P and I, or equivalent measurements.
2. Subtract dark current, background power, electronic offsets, and drift estimates.
3. Apply splitter, responsivity, efficiency, gain, temperature, and timing calibration.
4. Test the dark state before forming normalized coordinates.
5. Form the observation vector y and its uncertainty or covariance estimate.
6. Generate the permitted state codebook under the chosen binary or weighted constraints.
7. Compare y with candidate codepoints using Euclidean distance after whitening, Mahalanobis distance, or a validated likelihood model.
8. Return a state only when confidence and margin over the second candidate exceed defined thresholds; otherwise return undecodable or repeat the measurement.
9. Log calibration version, temperature, integration time, gain setting, and raw observations for reproducibility.

$$\text{decoded state} = \arg\min_s (y - \mu_s)^T \Sigma_s^{-1} (y - \mu_s)$$

8. Precision, state capacity, and code-distance analysis

8.1 Three distinct questions must not be conflated

Question	Quantity	What it answers
How many states exist?	M_{input}	Number of permitted transmitter configurations
How many bins can the receiver report?	M_{bins}	Information-theoretic output capacity before code geometry
How many states are reliably decodable?	$M_{decodable}$	States remaining distinct after mapping, quantisation, noise, drift, and rejection

$$M_{decodable} \leq \min(M_{input}, M_{bins})$$

The inequality is necessary but not sufficient. A codebook can have fewer states than bins and still collide because the valid means are irregularly placed.

8.2 Nominal decimal capacity

If a normalized measurement is confined to [0,1) and reported to d decimal places, it has at most 10^d nominal bins. For n binary spectral lanes, including the all-off state, there are 2^n input configurations. Therefore:

$$2^n <= 10^d$$

$$n <= \text{floor}(\log_2(10^d)) = \text{floor}(d \log_2(10))$$

For d = 3, the ceiling is floor(log_2 1000) = 9. This proves only that ten binary lanes cannot fit into 1000 bins. It does not prove that a given nine-lane log-prime codebook fits.

Decimal places d	Nominal bins 10^d	Binary upper bound floor(log2 10^d)
1	10	3
2	100	6
3	1,000	9
4	10,000	13
5	100,000	16
6	1,000,000	19

8.3 Exact codebook for unknown active count

For n binary primes, define the non-empty mean codebook:

$$C_n = \{ \mu(S) : \emptyset \neq S \subseteq \{1,\ldots,n\} \}$$

$$\mu(S) = B + (A/|S|) \sum_{i \in S} \log(p_i)$$

There are 2^n - 1 non-empty means plus the separately classified dark state. Exact uniqueness follows from Section 3.3. The global unscaled minimum separation is:

$$\delta_n = \min_{S \neq T} | m(S) - m(T) |$$

After affine scaling, the ideal coordinate separation is |A| delta_n. B has no effect on pairwise separation. The closest pair can involve different active counts because count is not assumed known.

8.4 Quantisation and rounding collisions

Let Q_d(x) denote rounding to d decimal places. A finite-precision codebook is collision-free only if:

$$Q_d(\mu(S)) \neq Q_d(\mu(T)) \text{ for every } S \neq T$$

The simple condition |A| delta_n > 10^-d is sufficient for favourable bin alignment but is not strictly necessary: two points less than one bin apart can round differently if a bin boundary lies between them. Conversely, a gap near one bin can still be unsafe once noise or drift is added. Therefore report both minimum continuous separation and actual quantised-bin collisions.

8.5 Worked enumeration: first n primes

The following results use p = 2,3,5,7,11,13,17,19,23, binary subsets, base-10 logarithms, A = 1, B = 0, and means across all non-empty subsets. “Unique at 3 d.p.” counts distinct rounded mean values, not robust hardware states.

n	Non-empty states	Minimum mean gap	Unique values at 3 d.p.	3 d.p. collisions	Minimum d.p. for unique rounding
1	1	n/a	1	0	0
2	3	0.088045630	3	0	1
3	7	0.007626248	7	0	2
4	15	0.007490806	15	0	2

5	31	0.001007370	31	0	3
6	63	0.000599033	63	0	3
7	127	0.000012032	117	10	5
8	255	0.000012032	208	47	5
9	511	0.000001318	343	168	6

Interpretation: With the first primes, six lanes happen to produce 63 distinct rounded means at 3 d.p., despite a minimum continuous gap below 0.001. Seven lanes do not: 127 values collapse to 117 rounded bins. Therefore the nominal nine-lane ceiling is not attained by this unoptimised codebook. For engineering reliability, even six lanes have inadequate margin if one measurement step is 0.001.

8.6 Four-prime worked example

For primes 2, 3, 5, and 7, the 15 non-empty subset means are shown below. The closest pair is the four-prime mean and the mean of {3,5}.

Active primes	Mean log10 coordinate
{2}	0.301030
{2,3}	0.389076
{3}	0.477121
{2,3,5}	0.492374
{2,5}	0.500000
{2,3,7}	0.541083
{2,7}	0.573064
{2,3,5,7}	0.580555
{3,5}	0.588046
{2,5,7}	0.615033
{3,7}	0.661110
{3,5,7}	0.673730
{5}	0.698970
{5,7}	0.772034
{7}	0.845098

$$\text{delta_4} = 0.588045630 - 0.580554824 = 0.007490806$$

At ideal 3-decimal rounding the pair becomes 0.588 and 0.581. The gap spans about 7.49 nominal 0.001 steps before noise, which is substantially safer than the five-prime minimum of about 1.01 steps.

8.7 Scaling by A and placement by B

Multiplying by A scales every mean and every gap. Adding B translates the codebook without changing any distance:

$$\text{mu_C}(S) = B + A\, m(S)$$

$$\text{Delta_C,min} = |A|\ \text{delta_n}$$

If the detector coordinate has usable span R, all scaled means must also fit:

$$|A| [\max_S m(S) - \min_S m(S)] <= R$$

Thus A cannot be increased without limit. Increasing A improves absolute spacing only by consuming more analogue range. The scale-selection problem is: maximize the minimum physical separation while keeping the full calibrated codebook inside the linear, unsaturated receiver range. B should centre the codebook or align it with a convenient operating region; B does not improve distinguishability.

$$A_{max} = R / [\max_S m(S) - \min_S m(S)]$$

For a selected prime set, the best affine-only normalized packing is obtained at the largest A allowed by range and linearity. Prime selection remains important because affine scaling cannot change the ratio of the minimum gap to total span.

$$\text{packing quality } \rho = \Delta_n / [\max_S m(S) - \min_S m(S)]$$

8.8 Noise-aware design margin

Let sigma be the RMS uncertainty of the calibrated scalar code coordinate, including detector noise, amplifier noise, source fluctuation, calibration residuals, temperature drift, spectral shift, and timing error. Let q_ADC be one converter step expressed in the same coordinate. A conservative scalar rule is:

$$\Delta_{min,physical} >= 6\sigma + 2 q_{ADC} + 2 \Delta_{drift}$$

The coefficients are design choices, not universal constants. Six sigma corresponds to three sigma allocated on each side of adjacent ideal states before additional quantisation and drift allowances. The correct criterion should be validated by measured distributions and target symbol-error rate.

For multidimensional observations, whiten the calibrated codepoints using their covariance and use the minimum Mahalanobis separation:

$$\Delta_{min,white} = \min_{s \neq t} \sqrt{(\mu_s-\mu_t)^T \Sigma^{-1} (\mu_s-\mu_t)}$$

Simulation should report confusion matrices and empirical symbol-error rates rather than infer reliability from decimal places alone.

8.9 State-selection and prime optimisation

Using the first n primes is a convenient demonstration, not necessarily the best finite-precision code. For a candidate prime set P of size n, define:

$$P^* = \arg\max_P \Delta_n(P) / \text{span}(P)$$

subject to optical bandwidth, manufacturable channel spacing, source power, filter bandwidth, detector response, and calibration constraints. Searching over prime sets may increase code distance, but the resulting primes must map to physically realisable frequencies through A log(p) + B. Optimisation must therefore include both arithmetic code distance and optical feasibility.

8.10 Required precision-analysis report

- Permitted prime set, logarithm base, A, B, and optical range.
- All valid codepoints or a reproducible generator.
- Continuous minimum, 1st percentile, median, and maximum nearest-neighbour distances.
- Identity and active counts of the nearest state pair.
- Rounded-bin collisions at each tested ADC precision.
- Noise, drift, calibration, and nonlinearity assumptions.
- Monte Carlo symbol and per-lane error rates.
- Rejection rate and confidence-margin distribution.

- Sensitivity to wavelength shift, source imbalance, detector responsivity, and temperature.
- Comparison of factorisation, scalar mean-codebook, and multidimensional statistical decoding.

9. Orthogonal frequency and amplitude extension

9.1 Eight binary spectral controls

Let b_i in {0,1}, i = 0,...,7 select eight calibrated spectral lanes. The transmitter has 2^8 = 256 spectral words, including the all-off word. The 255 non-empty patterns map to mean codepoints; the all-off word is detected by the dark-state test.

$$F = \sum_{i=0}^{7} b_i 2^i, \quad F \in \{0,\ldots,255\}$$

The existence of 256 words does not guarantee three-decimal decodability. For the first eight primes with A = 1 and B = 0, only 208 distinct values remain after three-decimal rounding. A practical eight-lane design therefore needs an optimised prime set, more precision, more physical range through A, a multidimensional receiver, or subdivision into smaller spectral blocks.

9.2 Eight binary-weighted amplitude controls

Let a_j in {0,1}, j = 0,...,7 drive additive optical power increments with weights 1,2,4,...,128. For mutually incoherent sources, phase-averaged signals, or deliberately combined optical powers, the total power scale is:

$$D = \sum_{j=0}^{7} a_j 2^j, \quad D \in \{0,\ldots,255\}$$

$$G(D) = G_0 + D\,\Delta G$$

The 256 amplitude levels are uniformly spaced in the intended calibrated gain coordinate. Coherent electric fields should not be described by simple power addition unless phase and interference are explicitly controlled.

9.3 Decoupling by common spectral scaling

To encode an amplitude word independently of the spectral word, the amplitude control should scale every active spectral lane by the same factor G(D):

$$\Phi_i = G(D)\, b_i \Phi_0$$

Then the normalized mean is independent of D:

$$\text{mean}(\nu) = \left[\sum_i G(D) b_i \Phi_0 \nu_i\right] / \left[\sum_i G(D) b_i \Phi_0\right]$$

$$\text{mean}(\nu) = \left[\sum_i b_i \nu_i\right] / \left[\sum_i b_i\right]$$

The aggregate intensity depends on D and the number of active spectral lanes. Therefore an amplitude decoder must either compensate for spectral occupancy, use a transmitter architecture in which total reference power is held independent of occupancy, or decode the full two-dimensional codebook jointly. A naive total-power reading is not automatically D alone.

Important coupling condition: If total power equals G(D) times the number of active spectral lanes, then different pairs (F,D) can share the same total power. Independence requires a known normalization, constant-power spectral coding, an additional reference, or joint calibrated decoding. The 65,536 count is a permitted input-space target, not automatic proof of an injective physical readout.

9.4 Theoretical two-observable state count

If F and D are independently selectable and the calibrated receiver maps each pair to a unique resolvable observation vector, then:

$$M_{input} = 256 \times 256 = 65{,}536 = 2^{16}$$

$$\text{state index } W_{16} = 256F + D$$

This is equivalent to a theoretical 16-bit optical symbol. The actual decodable count is:

$$M_{decodable} \leq \min(65{,}536, M_{bins}, M_{separated})$$

where $M_{separated}$ is the number of input pairs that remain distinct after calibration, quantisation, noise, and rejection criteria.

9.5 Receiver options for the two-dimensional code

Architecture	Observable 1	Observable 2	Qualification
Split-path aggregate receiver	Total calibrated power or charge	Normalized spectral/energy proxy	Needs wavelength-dependent calibration and occupancy treatment
Weighted filter pair	Broadband total response	Spectrally weighted response	Decodes a finite codebook; two integrals do not reconstruct an arbitrary spectrum
Spectral demultiplexer plus power receiver	Explicit spectral bins	Total or per-bin power	More detectors but simpler identifiability
Energy-resolving detector plus count channel	Event count	Energy distribution or total energy	Technology-dependent resolution and count rate

10. Experimental architecture

Source -> collimation -> spectral dispersion -> calibrated mask/modulator -> optional physical logic stage -> calibrated aggregate receiver. A two-path implementation uses a characterised splitter, synchronous acquisition, and measured wavelength-dependent transfer functions. A broadband LED can support masked spectral demonstrations when lane bands are separated and calibrated; coherent gates require appropriate coherent sources and phase stability.

11. Minimum calibration record

Record lane centre and bandwidth; single-lane P and I means, variances, and linearity; dark/background versus temperature and integration time; cross-talk; path-split transfer functions; amplitude-gain levels; codebook points; timing; and calibration version. The 8+8 mode additionally requires every intended amplitude level across representative spectral patterns to test separability and coupling.

12. Recorded software demonstration

The existing 12-lane result remains evidence of the ideal mathematical pipeline and mapped-lane Boolean convention. It does not establish physical optical NOR behaviour, P/I calibration, detector accuracy, error rate, or 65,536-state operation.

13. Software changes required

Replace uniform-voltage capacity claims with codebook enumeration. Add a binary subset-mean generator, exact uniqueness tests, rounded-bin collision reports, global nearest-neighbour analysis, prime-set optimisation, two-dimensional frequency/amplitude codebooks, occupancy-aware amplitude normalization, covariance-based decoding, dark-state handling, confidence rejection, and calibration versioning.

14. Supported claims and limitations

Supported: formal log-prime coordinates, exact binary subset-mean uniqueness, ideal prime-product recovery when total weight is known, software mapping and ideal simulated decoding, and proposed calibrated receiver architectures. Not yet supported: physical recovery of the full codebook, a fixed decimal-place capacity, a physical optical NOR, eight spectral bits at three decimal places, 65,536 physical 8+8 states, high-speed low-error operation, or patentability.

15. Detectorless intermediate processing extension

The aggregate decoder is principally an end-of-pipeline readout. Multiplexed lanes may remain optical through passive, nonlinear, interferometric, or magneto-photonic transformations. Any future stateful optical element must be evaluated for restoration, gain, fan-out, isolation, cross-talk, switching energy, loss, and cascadability.

16. Validation plan

Phase A: single-lane spectral and detector calibration. Phase B: enumerate and measure binary subset codebooks, beginning with four lanes and increasing only while margins remain adequate. Phase C: test prime-set optimisation and calibration transfer. Phase D: add binary-weighted common amplitude scaling and measure the full two-dimensional code geometry. Phase E: demonstrate physical logic only after the receiver is validated. For every phase publish raw data, scripts, uncertainty budgets, nearest-neighbour pairs, confusion matrices, and rejection rates.

17. Publication and licensing statement

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Appendix A - Conditions for exact and practical recovery

Case	Required conditions
Exact mean-codebook uniqueness	Distinct primes; binary subsets; non-empty state; exact arithmetic. Active count need not be known.
Direct prime-product factorisation	Known A, B, log base, and N or W; non-negative integer exponents; enough arithmetic accuracy.
Finite-precision scalar decoding	All permitted means map to unique bins with adequate

	noise and drift margin.
Two-observable decoding	Calibrated injective codebook in the observation plane, measured covariance, dark handling, and rejection logic.
8+8 extension	Independent control of spectral and amplitude words; common-scale or otherwise calibrated transmitter; receiver separates every permitted pair with margin.

Appendix B - Reproducible precision algorithm

For each n and candidate prime set:

- Enumerate masks from 1 to 2^n - 1.
- For each mask, calculate the arithmetic mean of the selected log-prime coordinates.
- Sort the means and calculate adjacent differences.
- Record the minimum gap and the two subsets producing it.
- Round every mean to each candidate decimal precision and count unique bins and collisions.
- Apply scaling A and offset B, then repeat in physical receiver units.
- Add noise and calibration perturbations by Monte Carlo simulation.
- Decode by the intended rule and report symbol error, lane error, and rejection rate.

Complexity: Exact binary codebook enumeration is O(2^n) in state count. It is trivial for n <= 12, manageable for moderate n, and eventually requires structured search, pruning, or coding-theory methods.

Appendix C - Recommended terminology

Avoid	Use instead
Three decimal places supports nine frequencies	Nine is the 1000-bin upper bound; actual support depends on code distance and collisions
Photodiode voltage equals mean photon energy	Calibrated observables are transformed into a mean-code estimate under stated assumptions
Amplitude adds	Optical powers add for incoherent or phase-averaged components; coherent fields may interfere
65,536 demonstrated states	65,536 permitted 8+8 input pairs in the theoretical two-observable extension
Exact hardware tolerance	Measured code-distance margin and empirical error/rejection rate
4096 equally spaced states	4096 permitted states with non-uniform calibrated code geometry

Appendix D - Version 2.1 concise conclusions

- Prime-derived logarithmic subset means are exactly unique for distinct non-empty binary subsets, even when active count is unknown.
- Direct product factorisation still requires active count or total integer weight.
- Three decimal places provide at most 1000 scalar bins and therefore at most nine binary variables in principle, but the unoptimised first-prime mean codebook collides by seven lanes at 3 d.p.

- For the first primes, four lanes have substantial ideal separation, five are marginal at a 0.001 step, six are collision-free only because of bin alignment, and seven collide.
- Practical precision must be specified through calibrated code distance, quantisation, noise, drift, confidence, and rejection rate, not decimal places alone.
- Eight spectral bits plus eight independently encoded amplitude bits define 65,536 permitted input pairs, but physical recovery requires an injective, calibrated, noise-tolerant two-observable mapping.
- The next decisive experiment is a measured four-lane codebook followed by controlled scaling to five and six lanes, then a two-dimensional amplitude-spectral prototype.